

MARK WU

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TECHNICAL SKILLS

Languages: Python, TypeScript, SQL · **Working knowledge:** Rust
Web & Backend: React, Tailwind CSS, FastAPI, PostgreSQL, Tauri
ML & Computer Vision: scikit-learn, HRNet, YOLO, ByteTrack
Embedded & CAD: ESP32, Pixhawk, ArduPilot, MAVLink, SolidWorks, Fusion 360
Tools: Git · **Spoken:** English, Mandarin (bilingual)

EDUCATION

University of California, Merced <i>B.S. Mechanical Engineering, Aerospace Engineering emphasis</i>	Merced, CA <i>Expected 2030</i>
Diamond Bar High School <i>AP Computer Science A, AP Calculus BC, AP Physics C: Mechanics, PLTW</i>	Diamond Bar, CA <i>June 2026</i>

EXPERIENCE

FRC Team Sprocket 3473 <i>CAD Subteam; Scouting Software Subteam Founder (Alumnus)</i>	Diamond Bar, CA <i>2024 – 2026</i>
– Founded the scouting software subteam after the legacy app became a competitive liability; served as de facto technical lead of the subteam, directing feature decisions and mentoring 3 contributing developers while personally writing the majority of the codebase – Secured a dedicated subteam sponsor (HumanSignal) and managed the infrastructure budget across Neon, Fly.io, and domain hosting – Deployed live at 3 FRC competitions, including both 2026 CA district events, capturing 411 robot-match records across 73 teams – Team awards during tenure: Regional FIRST Impact (2025), Engineering Inspiration (2026), Worlds (2024, 2025)	
Team Infernope <i>Combat Robotics, Design and Fabrication</i>	ShenZhen, CN <i>2020 – 2024</i>
– Designed and built twelve combat robots over three years (chassis, drivetrain, weapons, in-match repair); placed 1st in the 2024 end-of-year tournament	
SoCal FTC Qualifiers <i>FIRST Technical Advisor (Volunteer)</i>	Diamond Bar, CA <i>January 2026</i>
– Ran one of two competition fields as FTA at two qualifier events, diagnosing hardware, software, and robot control failures live during match play	

PROJECTS

SprocketStats <i>TypeScript, React, Python, FastAPI, PostgreSQL, scikit-learn, HRNet</i>	2025 – Present
– Built a match win prediction pipeline (Random Forest, K-Means) scoring ~87% across ~90 late-qualification and playoff matches, statistically indistinguishable from the Statbotics EPA baseline at 95% confidence in a proportions test – Developing v3, a computer-vision rebuild of data collection: homography field mapping, HRNet pose estimation, and YOLO + ByteTrack detection and tracking; currently standalone modules pending integration – Extending the platform beyond scouting into team operations: building a Tauri sign-in kiosk (in progress) against member IDs, either RFID tags or 8-digit checksummed codes, administered by the scouting subteam I continue to mentor	
Caelifer, Coaxial EDF VTOL <i>SolidWorks, Pixhawk 6C, ArduCopter</i>	2026 – Present
– Tube-launchable VTOL targeting multirotor-class hover with better cruise efficiency: coaxial contra-rotating EDFs in a 100 mm finless tube, pitch and yaw from servo-actuated exhaust vanes – Bench-testing the propulsion and avionics stack ahead of first hover test: coaxial 70 mm EDF pair on a 6S pack, 180 A peak draw (~4.0 kW)	
Project Aetherius, Fixed-Wing UAV <i>Tauri, TypeScript, Rust, MAVLink, ArduPilot, Pixhawk 6X</i>	2025 – Present
– Built a custom Tauri ground control station (3rd revision) for ArduPlane and ArduCopter: guided firmware installation, full vehicle calibration, failsafe and servo setup, mission planning, and an embedded Lua IDE – Implemented the ArduPilot layer over Tauri IPC for parameters, telemetry, and commands; used it on real hardware to flash firmware, calibrate, run pre-arm checks, arm, and switch modes during the airframe's first flight – Built a 2m twin-boom airframe (4th revision) around a Pixhawk 6X; flew it for the first time in August 2026, two self-propelled takeoffs and roughly ten seconds of airtime combined, flown manually on RC	